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RESEARCH

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Addis Ababa online home rental management system, Ethiopia

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Abstract

The purpose of this project is to plan and create an online home rental management system in Ethiopia's capital, Addis Ababa. This system aims to streamline all tasks associated with the proper administration of rental properties through an online platform. Its primary objective is to reduce the time required to complete rental-related tasks. The current manual record-keeping system is slated for replacement to enhance efficiency in calculations and data storage. The system's database backend operates on MySQL server, with C# and PHP serving as the front-end programming languages. Periodic purging of the database is scheduled, ensuring the system's ability to handle daily operations effectively. This approach aims to minimize manual calculations and labor-intensive tasks while maintaining regularity in operations. Managing data related to various aspects such as products, employees, managers, and clients is a key component of this system. Furthermore, this newly implemented system boasts an excellent database structure that facilitates collaboration among geographically dispersed users. It features a flexible and user-friendly interface for quick record retrieval and enables tenants to view available properties within the system.

Keywords: Home renting, House management, Addis Ababa, Ethiopia

Introduction

In addition to being a fundamental human right, housing serves as a private, dynamic space for a variety of social interactions and activities as well as a place of refuge [1]. Countries all over the world provide a variety of rental housing options. Ballesteros [2] states that there are differences in rental housing in size, construction, quality, ownership, rent, kind of contract, and profitability. There are many different kinds of rental housing. For example, rental housing is owned by both the public and private sectors and is available in a range of configurations. It can be rented by a variety of property owners and can be as opulent as penthouses [3].Thus, the primary characteristic of rental housing is diversity.

There are two primary categories of rental housing: private and public. Individuals and commercial entities supply private housing. This covers housing that firms provide to their staff and temporary housing arrangements made between friends or family. Most of these lodging establishments do not have a local business license [2]. It can be difficult to tell if a rental property is in the legitimate market because several affluent or middle-class rental properties lack a company registration for rentals. Thus, it is difficult to distinguish between formal and informal rental housing [2]. In contrast, the government owns public rental homes.

A vital housing option that can also be a financial decision is rental housing, particularly for individuals who cannot purchase a home. Additionally, people with bad credit records, little home renovation reserves, or those particularly vulnerable to income fluctuations should consider this option [4]. Sixty percent of Addis Ababa City's population, according to [5], is in rented housing. However, [6] population and housing census found that the rental sector provided 58.8% of the housing in Addis Ababa, with the private sector accounting for 37% of the total and the public sector, primarily the Kebele, providing the remaining portion [6].

Ethiopia is facing a severe shortage of rental accommodation in its cities. There is a serious lack of reasonably priced private rental housing, especially in Addis Ababa [3]. Belsky and Drew [4] report that there is a significant discrepancy between Addis Ababa's supply and demand for residential properties. Due to the government's shortage of reasonably priced rental housing and the persistently high demand for residential land, municipal workers in the public sector have restricted access to affordable private rental housing. Since many public servants stay in private rental homes, it is thought that finding accessible and inexpensive rental accommodation is the main issue facing Addis Ababa citizens.

Studies conducted in Addis Ababa City, however, have not sufficiently addressed this problem to date. Therefore, it is crucial to determine how economical and accessible private rental housing is for households headed by public sector employees. To achieve this, the paper makes an effort to provide a thorough explanation of private rental housing practices and how they affect public officials' access to policies in Ethiopia, specifically as it relates to the Addis Ababa City Administration. Since neither the rental nor the homeowners have access to the information that brokers play with, there are a lot of brokers in the neighborhood that demand large commissions from both parties to rent a house. This is taken into account in the system design to improve communication among homeowners.

Literature review

A home is a place of habitation that is used as a permanent or semi-permanent place of residence by a person, family, household, or numerous families within a tribe. The landlord lets someone use a room in their home that is always available for short-term occupancy by paying the renters [7]. This room is only used as a hotel, motel, or other similar establishment; it is not a residential unit.

A place to call home is among the essentials for human beings. These days, someone who does not have a place to reside can pay rent to live in someone else's house. In this instance, a contract outlining specific guidelines and circumstances around payment should be signed by the landlord and tenant. A residence that is rented out is one from which the owner gets paid by the tenant(s) for utilizing or occupying the space. There are two types of rental properties: residential and commercial. Certain tax deductions, such as mortgage interest and depreciation, may be available to rental property owners [8].

The new digital platform called the House Rental Management System is revolutionizing the management of rental properties. It provides a quick and convenient alternative for landlords and tenants to manage a variety of rental property issues. Tenant administration, tracking of rental listings and availability, automatic rent collection, handling of maintenance requests, financial tracking and reporting, and secure document management are some of the key features [9].

Statement of the problem

The problems found in the current manual home renting system:

- Difficult to searching home.
- Extra money like transportation, pay for the third party to get the certain apartment.
- The method requires additional human labor.
- The user is unable to obtain information about their home when needed.
- Time consumption
- The landlord cannot be present when the tenant wants to rent and release the home
- The intricacy of the payment system, when something breaks in the rental unit, putting the tenant's health and safety in jeopardy or endangering the building or property until repairs can be completed, an emergency repair is necessary.

Features of the proposed system

- The proposed system has an interactive user interface and can avoid most of the issues with the current system.
- It has prescheduled user input text boxes to fill in home records, as well as an advanced and regular search function that allows users to search for a specific home record using different refining criteria. This facilitates quick and simple data access.
- The system uses relational databases to prevent data loss and efficiently aggregates residential information into many categories. For instance, convenient data access is implemented and data redundancy is avoided by utilizing a single primary key to relate the tenant and property owner's virtual bank tables.

The system uploads images of the house along with the information that is accessible, and the user can view the photos of the house that they require.

System architecture

System architecture is the way a system is structured, including its components, their relationships, and how they work together. Some other types of system architecture. Include:

Event-driven architecture (EDA)

A method for designing software systems that enables efficient communication between components or services through events.

Three-tier architecture

A popular implementation of a multi-tier architecture that consists of a presentation tier, logic tier, and data tier. This system focused on the online Home rental management system's design in this chapter. It assists us in understanding the specifications of the architecture, parts, modules, interfaces, and data needed a system to meet predetermined standards. It suggests a methodical approach to the system's design. This section covers the various class-type architecture types as well as the many system-modeling approaches that are applied to the system's implementation.

Current system

The current system in the sense the system file handling system is manual and this causes loss of data, hard-to-retrieve old data, high data, and information redundancy very exhaustive file handling (because of vast numbers of manually handled data), and inconstancy on data retrieval, update and remove.

The new developed system

The effective and open management of rental properties has advanced significantly thanks to house rental management system [10]. Once the user finds the place, which fulfills his needs, he can directly contact the owner. There will be one more available option for the owner who can also add new rental homes as per availability. Therefore, the webpage will act as a broker between the owner and the tenant with no extra cost and transparency. Both the parties will be in direct contact and this will help in saving time and money for the users [11]. UTHM House Rental Application (UTHM HR App) is a mobile-based application that will act as a third-party software mainly to help ease the business processes of students making rent in Parit Raja [12]. The project will adapt with throwaway prototyping methodology. An accurate prediction of house prices is a fundamental requirement for various sectors, including real estate and mortgage lending [13]. This newly developed system uses a desktop application. This system:

- Enables the manager to manage the system easily
- Effective time handling
- Easy and free from complexity for system users
- It protects data illegally accessed by unauthorized users.
- Good data handling system
- Improving the existing system
- Creating preferable performance

Class type architecture

Interface layer

This layer is where people and machines communicate with one another. The purpose of this interaction is for the user to effectively operate and control the machine, and for the machine to provide feedback to help the operator make operational decisions. Two varieties of interface layers exist.

Layer of business/domain

This section of the software encodes the business rules that apply in the actual world and specifies how data can be created, viewed, saved, and altered. The deletion, addition, and update functions in this system are examples of business activities, which are focused on, in our business domain. For instance, it configures the interactions between business objects.

Layer of persistence

It involves creating, reading, updating, and deleting persistent objects as well as transactions using a data source and a session that is associated at runtime. The in our system database contains a variety of tables that are either directly or indirectly related to one another.

Layer of systems

The programs, platforms, application servers, containers, runtime environments, packaged applications, virtual time, and all other elements described in this layer together with the runtime and development framework. PHP, MySQL, and C# used in our system.

Breakdown of the system

System decomposition involves disassembling a system into its constituent subsystems, examining each one independently, and then reassembling them into the system as a whole. Figure 1 shows that they divide apart the duties that the code and application must perform to provide value to the user. The layers do not consider the physical locations of components; instead, they are concerned with the logical division of components and functions.

If there is a change in one layer we can change without changing others and it is good for maintaining.

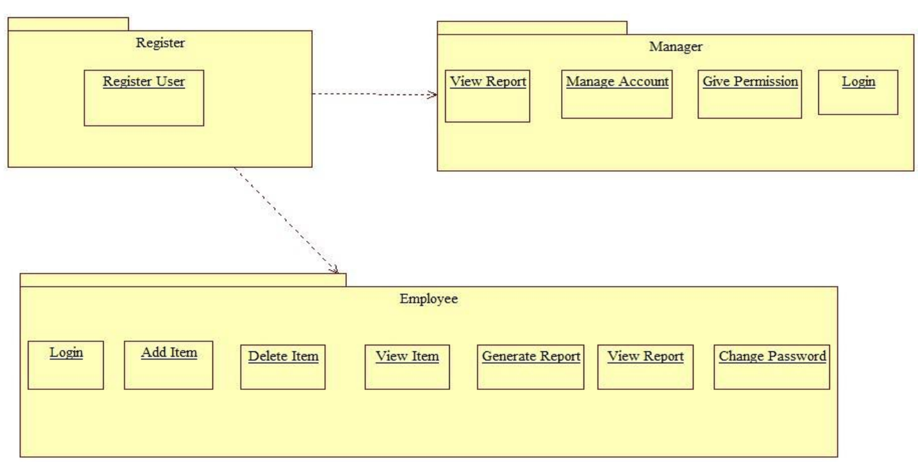

Fig. 1 System decomposition diagram

System flow chart

Figure 2 shows that the way this system operates is that each time a user wants to access it; they start by entering their username and password, going to the main page, and then selecting the desired sub form. Change Password Page enables all system users to change their password only under the main forms of the Manager Form.

- View Report Page
- Manage Account Page
- Allow Permission Page
- Send Command Page
- Manage Account(Add, Delete, Update Accounts) Page
- Invoice Page
- Proposed Item Report Page
- View Reports Page
- Company Information Page
- Change Password Page

Deployment diagram

The deployment model as shown in Fig. 3 displays the processing nodes' run-time configuration and the components that operate on them statically. The system's hardware, the software that is installed on it, and the middleware that links the various computers are all displayed in the deployment model.

Access control and security

The selective limitation of access to one location over another is the main issue of this unit. Accessing something can also imply entering, utilizing, or devouring it. Two similar access control mechanisms are locks and login. Our system will require a login. Access to the system must be controlled by different authentications. This authentication has a username and password. This username and password have their access to that user's page.

- The employee can only access their page that has different functions that are appropriate for their purpose. In addition, they can change their password but cannot create and delete accounts.
- The manager can access almost all of the data. And can add, delete… the account

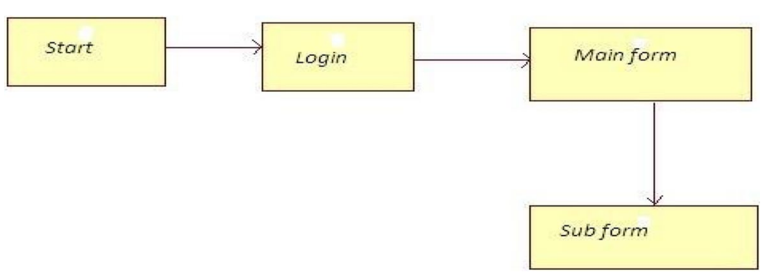

Fig. 2 System flow diagrams

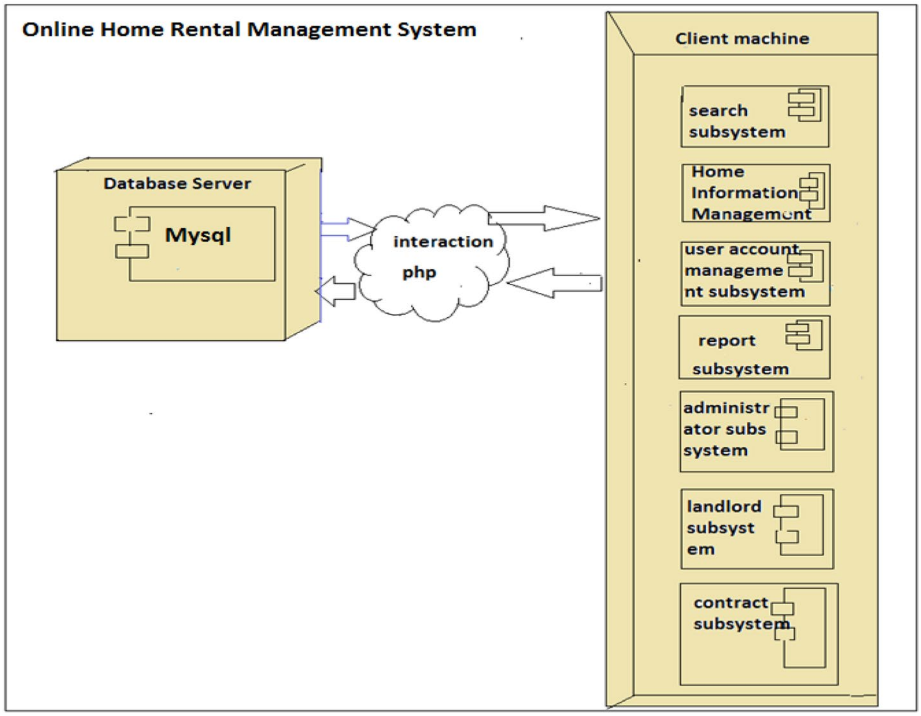

Fig. 3 Deployment Diagram of Online Home Rental Management Systems

RDBMS

An open-source, free relational database management system (RDBMS) example for use with relational databases is called My SQL (Structured Query Language) Server. Data are arranged in a table or relations using a relational database. There are rows and columns in a table. Another name for a row is a record or tuple. Another name for a column is an attribute or field. A spreadsheet and a database table are comparable. A relational database can store enormous volumes of data and retrieve specific data with ease thanks to the relationships that can be formed between tables.

Primary Key: A table cannot have duplicate entries in the relational model, as this would lead to retrieval ambiguities. Every table needs to have a column (or group of columns) named Primary to guarantee uniqueness.

Persistent data modeling

When a data structure is adjusted, it always maintains its previous version, which is known as a persistent data structure. The persistence layer creates, reads, updates, and deletes persistent objects by utilizing a session to correlate mapping metadata with a data source at run time. The purpose of persistence models, also known as data models or entity-relationship (ER) models, is to explain to users and other developers how a database—typically a relational database—is designed. Figure 4 shows the conversion of the class diagram into a tabular format using the mapping mechanisms. The database schema makes use of persistence. The conceptual similarity between

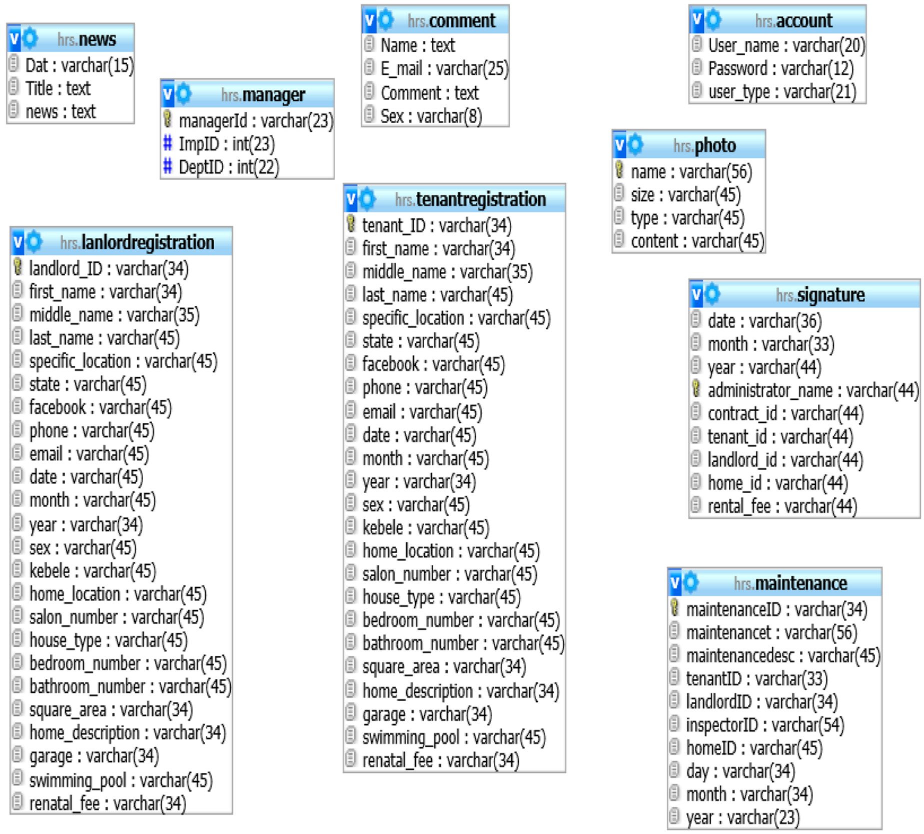

Fig. 4 ER Diagram

data entities and related database tables, as well as the fact that characteristics are the same, make persistence models strong.

Table 1 shows the necessary field or information to be required when registering new house available in to the system. The interface and database designed were fit with this necessary information to be fulfilled. This table was added as a sample where each of the entities has similar information in database.

Figure 5 shows the design of the database to produce physical and logical models of designs for the proposed database system.

User interface design

The design of computers, software, and webpages with an emphasis on user interaction and experience is known as user interface design as shown in Fig. 6. Making user interactions as simple as possible while still achieving user objectives is the aim of user interface design. A well-designed user interface makes it easier to complete the task without calling undue attention to it. To construct a system, the design process must strike a balance between visual components and technological functionality.

The User Interface, or UI, is the area where software and people interact. The user's efficient usage and control of the software is the aim of this interaction. Users can manipulate a system through its input, and the system can display the results of their manipulations through its output. For example in this system, there is a login

Table 1 House registration

No	Field	Type	Null	Description
1	LandlordID	Int(14)	No	Primary key
2	First name	Varchar(21)	No	landlord first name
3	Middle name	Varchar(21)	No	landlord middle name
4	Last name	Varchar(21)	No	Landlord last name
5	Specific_location	Varchar(45)	No	The place where the landlord is found
6	State/region	Varchar(34)	No	The state or the region where the landlord lives
7	Facebook account	Varchar(37)	Yes	Facebook username
8	Phone_number	Int(14)	N o	The landlord's phone number
9	Email	Varchar(37)	No	The email address of the landlord
10	Date	Varchar(50)	No	The landlord's birth date
11	Month	Varchar(37)	No	The date of the month
12	Year	Varchar(15)	No	The year when the landlord borne
13	Sex	Varchar(10)	No	The sex of the landlord
14	Kebele	Varchar(37)	No	The landlord kebele
15	Home_location	Varchar(37)	No	The location of where the home is found
16	Salon_number	Int(14)	No	Number of salons that are within the rented house
17	House_type	Int(14)	No	The type of house
18	Bedroom_number	Int(14)	No	The number of bedrooms that house has
19	Bathroom_number	Int(14)	No	The number of bathrooms that house has
20	Square_area	Varchar(37)	No	The area of the house
21	Home_description	Varchar(37)	No	A detailed description of the house to be rented
22	Garage	Varchar(37)	No	The availability of the garage for certain house
23	Swimming_pool	Varchar(37)	No	The existence of the swimming pool within the house
24	Rental_fee	Varchar(37)	No	The amount of payment for a rented house

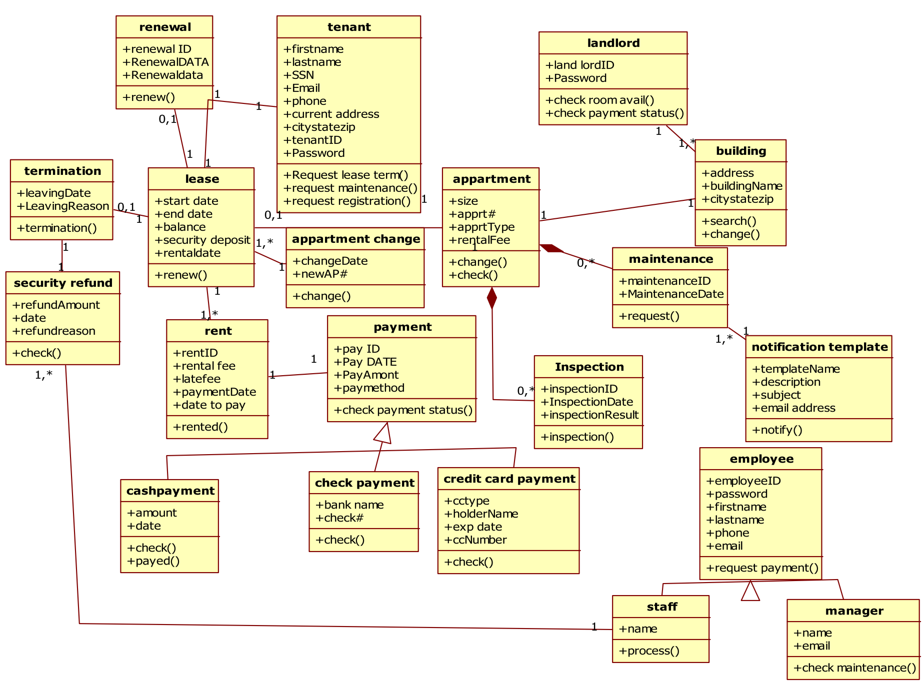

Fig. 5 Database Name: Online home rental management system

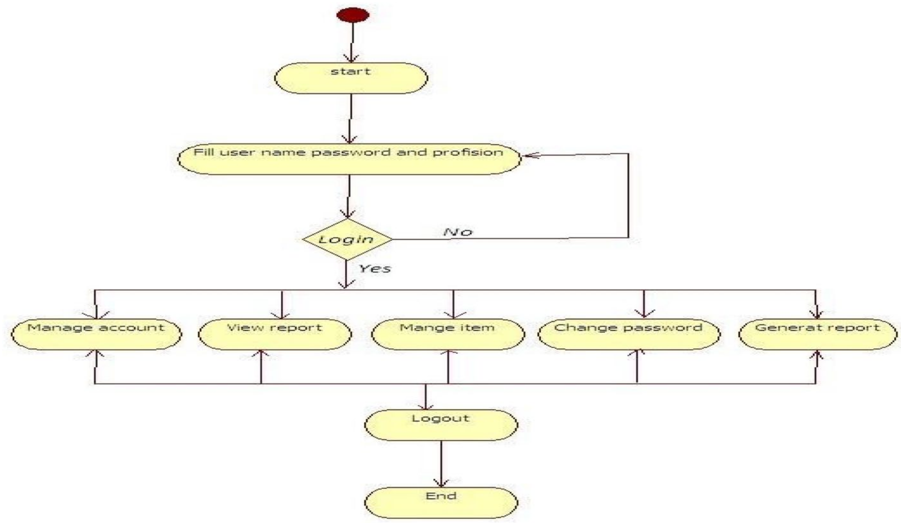

Fig. 6 System flow diagram

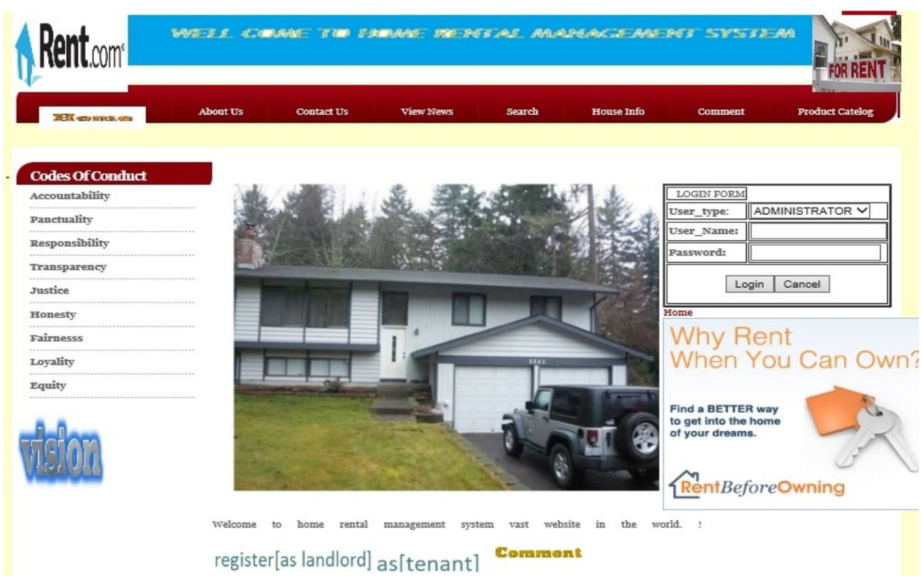

Fig. 7 Login Interface

that is used to interact with the system. The user login shows a user page and the manager login shows the manager page after accepting the correct username and password as shown in Figs. 7, 8, 9 and 10. In this system, there are the following interfaces.

Object design

We go over how to plan an object-interaction system in this chapter to address our software challenge. Encapsulated data and methods gathered to form an entity make up an object.

Fig. 8 Landlord registration form

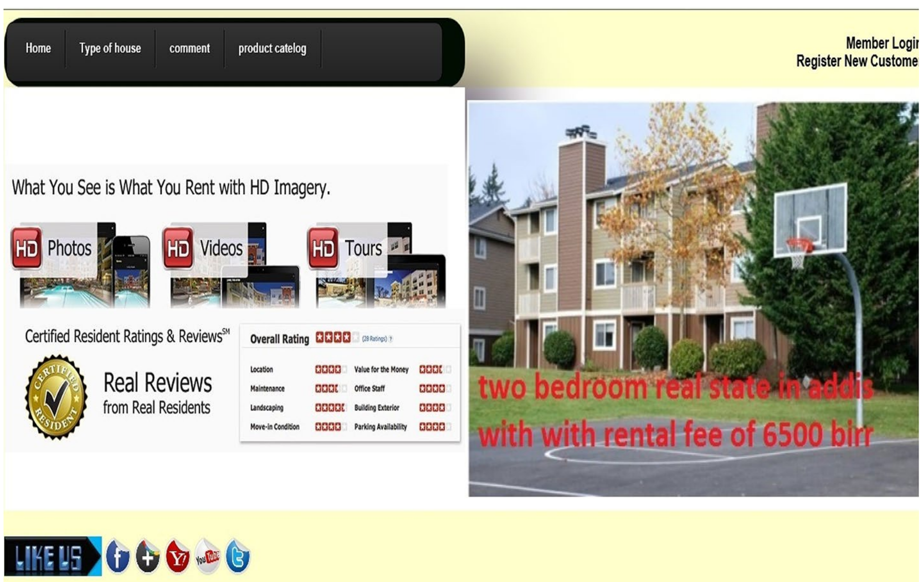

Fig. 9 House Information page

Enhanced class diagram

By outlining the many classes and objects inside the system as well as their relationships, a class diagram gives a general overview of the target system. The above objects' purposes are summarized in Table 2.

Visibilities

The visibility of the object is a measure of the distance at which an object can separate.

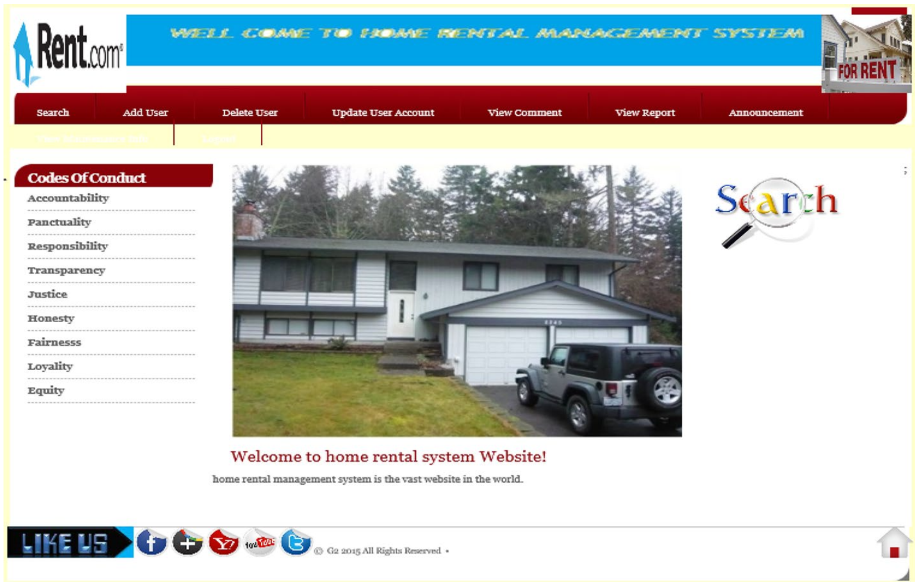

Fig. 10 House Information page

Table 2 Objects in class diagram

Class name	Purpose
Login	It is used to access the software appropriately. By accepting the appropriate username and password
Item	It is used to get different useful information about the specific house
Category	It is used to get different useful information about the specific category
Report	It used to send reports or accept reports
Worker	It gives different information about the house
Customer	It gives different information about the customer

- *Private*. The field or method in a private operation will only be accessible to the current class (Fig. 11).
- Safeguarded. The property or method in a protected operation will only be accessible to the current class and its subclasses, occasionally even the same package classes.
- The public. Any class can invoke the method or refer to the field in a public operation.

Implementing and testing

Testing

Any activity intended to assess a program or system's capability or attribute and ascertain whether it achieves the desired outcome is known as software testing. It is the act of running a system or program to identify errors. The process of validating and confirming that a program: • Complies with the specifications that informed its design and development, Function as anticipated; Have the same features when implemented; validate and/or test the process of running a program or piece of software to identify and check for errors and then fix (correct) them.

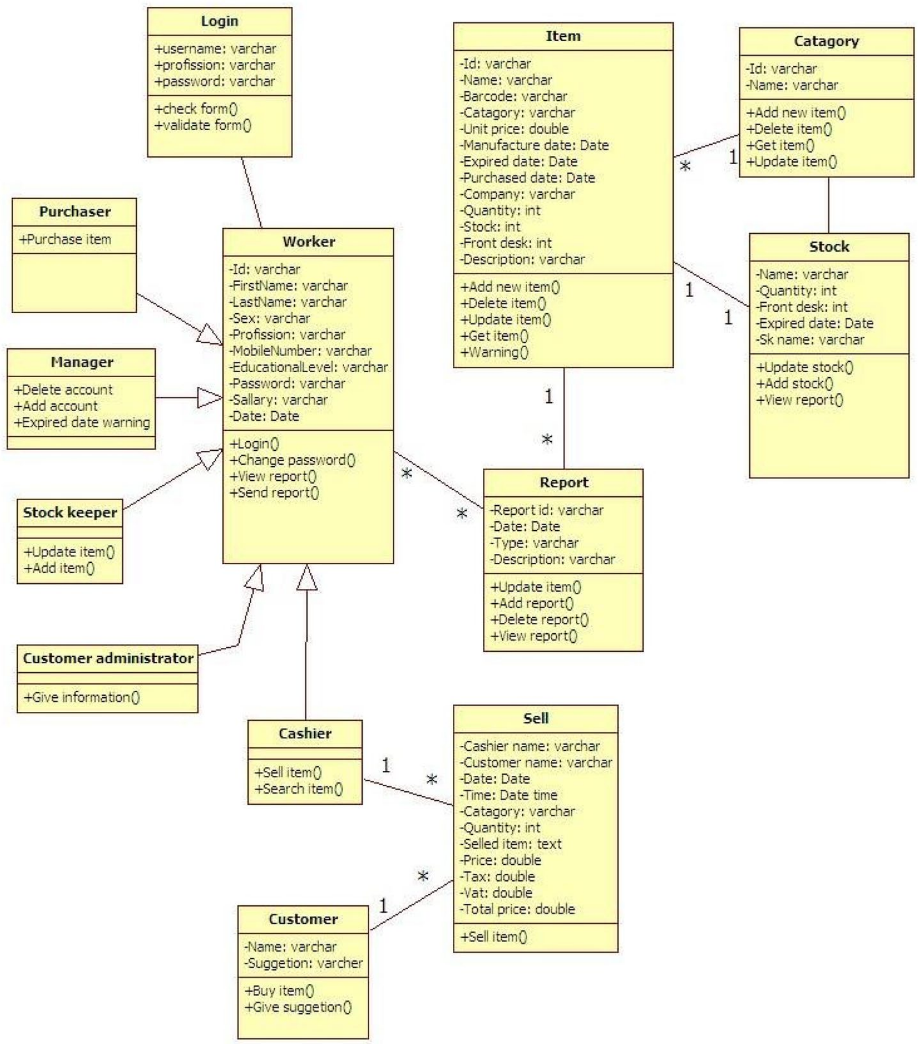

Fig. 11 Revised class diagram

Testing cannot show the absence of an error. There are two common testing approaches or techniques:

Black box testing

- A technique for testing the input and output of the system
- Is behavioral testing
- Is the test for functional requirement
- It looks for mistakes in the following classifications.

  An incomplete or inaccurate function
  Errors in the interface
  Access to other databases is incorrect
  Misconduct or performance issues

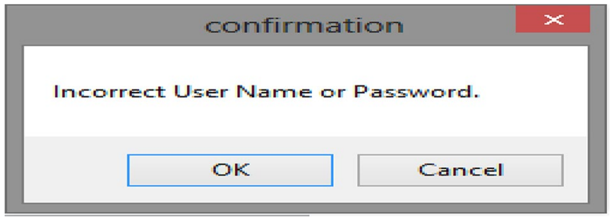

Fig. 12 Wrong password or user name confirmation message

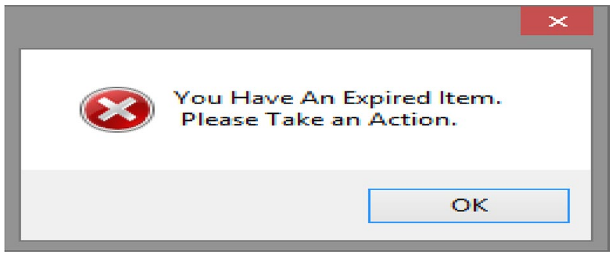

Fig. 13 Expiry date warning message

Errors in initialization and termination

White box testing

- Testing for internal procedures or data structures of the software
- It includes testing of

Sequential statement
Conditional structures
Functions and their calls

Features under testing

Unit testing

Individual software system unit is tested at the unit testing stage of the software testing process. The tiniest software component could be tested. It typically has one or more inputs and one or more outputs. As shown in Fig. 12, the system will indicate if someone enters the incorrect username or password.

As shown in Fig. 13, if the item is expired, the system will display this message.

If there is, a shortage of items in the stock the system will display this message shown in Fig. 14.

Integration testing

The stage of software testing known as integration testing is when separate software components are put together and tested collectively. It uses unit-tested modules as input,

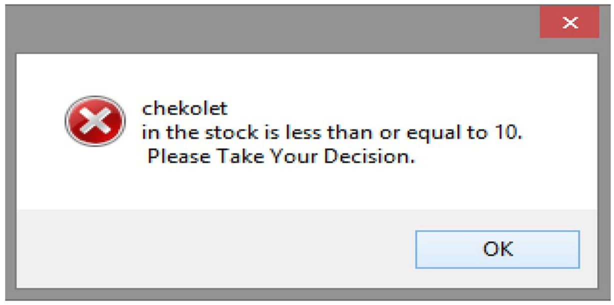

Fig. 14 Item depletion message

aggregates them into bigger groups, applies tests specified in an integration test plan to those groups, and outputs an integrated system that is prepared for system testing.

System evaluation

Software systems are tested as a whole, integrated unit to see if they meet the criteria as stated. This is known as system testing. System testing, a more constrained kind of testing, looks for flaws in the system as a whole.

Acceptance examination

- Testing to see if specifications or contracts are being followed is called acceptance testing.
- C#(Front end), PHP
- MySQL (Back end)

Specifications for testing cases

For the project to achieve the necessary functionality, it must pass a test case suit. New test cases verify the input description and compare the observed result with the expected output while existing test cases verify the previous capabilities. These are many requirements for test cases.

Testing for security: By attempting to access a system that is not authorized, security testing was carried out. For instance, a tester will attempt to enter the system if they do not have a working account and password.

Performance testing Evaluates the system's functionality across a variety of potential situations. To do this test, the system is configured in several environments, including various operating systems, networks, and hardware setups.

When the operating system test was done, the system operated perfectly for window versions greater than window XP. When it was tested for different hardware configurations systems response time was almost the same for Pentium 4 and above processors.

Conclusion

The software takes care of all the requirements of an average online home rental management system and is capable of providing easy and effective storage of information related to items that are in the online home rental management system. This project has shown documentation of an online home rental management system located in Addis Ababa. It issued to solve the home rental problem related to the data handling process. Problems have their effects on home rental. These problems are listed as follows:

- The problem with organizational structure: This problem will result in poor communication, delayed information exchange, and disorganized management of the home rental management system.
- Problem on file handling system. Data could be misplaced due to human error or in case of natural disaster confusion, physical damage, data loss, inefficiency, inaccuracy, and lack of internal control. This causes loss of data, hard-to-retrieve old data, security gaps, high data, and information redundancy very exhaustive file handling (because of a vast number of manually handled data), and inconstancy on data retrieval, updating and removal, and hard to manage resources.
- The system will try to overcome the problems by creating better performance by
- Effective time handling
- Good data handling system
- Good management system

The general objective of the system is to create a user-friendly & automated online home rental management system.

Maintenance

Throughout a system's operation, maintenance refers to updates and error correction. A system review is required to identify and fix the flaws are discovered in the system. A system evaluation can also tell the developer of the system's complete functionality, necessary modifications, and extra requirements. If a major modification is required, a new project must be initiated and carried out through the whole life cycle.

Conclusion and recommendation

We believe that this project has great acceptance by the users. Because it can solve some of the problems of the current online home rental management system. This project does not work on an online home rental management system including.

- Online Renting
- Online ordering
- Online Connection with suppliers

However, the customer can suggest the home rental management system to the customer administrator by presenting it to the property owner. The next team of other projects does this system into an online system and makes it so advanced. I recommend next researchers who would like to do online home rental management systems should include the above thing in their project.

In addition, these Addis Ababa administrations should give greater attention to the future modification and implementation of this project. They need to understand any system before trying to implement it because a great weakness was observed in understanding the current system, which they are using now.

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Consent for publication
This manuscript's author gave the Journal of Electrical Systems and Information Technology permission to publish it.

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